

Course 6044C

Semester VI

Theory

4 Credits

Industrial Automation

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6044C as Industrial Automation in Semester VI for Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Industrial Automation and Control course and polytechnic automation curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Automation designs, PLC logic, SCADA configurations and industrial networks must be based on approved engineering plans, certified hardware data, current safety standards and competent professional review.

Verified source note: Verified sources support automation hierarchy, PLC programming, SCADA, and DCS concepts. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Industrial Automation studies the systems and technologies used to automate industrial processes and machinery. It develops the ability to analyze automation hierarchies, evaluate Programmable Logic Controllers (PLC) and SCADA systems, understand industrial communication protocols, and coordinate automated solutions while respecting the technical and safety boundaries of electronic and automation engineering.

Malayalam support: □ handbook □□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□□.

Prerequisite foundation

Digital electronics, microcontrollers, control systems and basic industrial electronics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the hierarchy, components and benefits of industrial automation systems.
2. Apply Programmable Logic Controller (PLC) architecture and ladder logic programming.
3. Analyze the features and operation of SCADA and Human Machine Interface (HMI) systems.
4. Explain the systematic design of Distributed Control Systems (DCS) and industrial communication protocols.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO കണ്ടെത്തുക exam-ൽ ഉപയോഗിക്കുക വിശദീകരിക്കുക നിർവ്വഹിക്കുക വിശദീകരിക്കുക. Define, explain, apply ഉപയോഗിക്കുക action words ഉപയോഗിക്കുക.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the fundamentals and hierarchy of industrial automation systems.	Understand
CO2	Develop ladder logic programs for PLC-based automation tasks.	Apply
CO3	Analyze the operation of SCADA and HMI in industrial processes.	Apply
CO4	Explain DCS architecture and industrial communication standards.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Automation Fundamentals and Sensors	Definition and hierarchy of industrial automation (Automation Pyramid); Fixed, Programmable, and Flexible automation; Industrial sensors: Proximity, LVDT, Pressure, Flow; Actuators: Pneumatic and Hydraulic.	Approx. 14 hours	CO1	Module 1
2	Programmable Logic Controllers (PLC)	PLC Architecture: CPU, Memory, I/O modules; Scan cycle and operating principle; Programming languages: Ladder logic, FBD; Logic gates, Timers, Counters in ladder logic.	Approx. 16 hours	CO2	Module 2
3	SCADA and Human Machine Interface (HMI)	SCADA architecture and features: Data acquisition, GUI, Alarms, Trends, Logging; HMI design and operation; Interfacing SCADA with PLC; Remote monitoring basics.	Approx. 16 hours	CO3	Module 3
4	DCS and Communication Protocols	Distributed Control Systems (DCS) architecture; PLC vs DCS; Industrial communication protocols: Modbus, Profibus, CAN, Ethernet/IP; Industry 4.0 and IIoT overview.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Automation Fundamentals and Sensors

Definition and hierarchy of industrial automation (Automation Pyramid); Fixed, Programmable, and Flexible automation; Industrial sensors: Proximity, LVDT, Pressure, Flow; Actuators: Pneumatic and Hydraulic.

CO1 · Approx. 14 hours

Programmable Logic Controllers (PLC)

PLC Architecture: CPU, Memory, I/O modules; Scan cycle and operating principle; Programming languages: Ladder logic, FBD; Logic gates, Timers, Counters in ladder logic.

CO2 · Approx. 16 hours

SCADA and Human Machine Interface (HMI)

SCADA architecture and features: Data acquisition, GUI, Alarms, Trends, Logging; HMI design and operation; Interfacing SCADA with PLC; Remote monitoring basics.

CO3 · Approx. 16 hours

DCS and Communication Protocols

Distributed Control Systems (DCS) architecture; PLC vs DCS; Industrial communication protocols: Modbus, Profibus, CAN, Ethernet/IP; Industry 4.0 and IIoT overview.

CO4 · Approx. 14 hours

MODULE 1

Automation Fundamentals and Sensors

Definition and hierarchy of industrial automation (Automation Pyramid); Fixed, Programmable, and Flexible automation; Industrial sensors: Proximity, LVDT, Pressure, Flow; Actuators: Pneumatic and Hydraulic.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

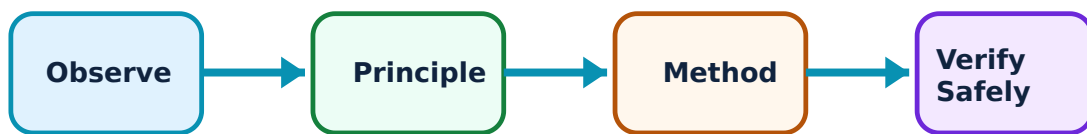
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Automation Fundamentals and Sensors: identify the system, state its principle, follow the correct method and verify the result safely.

1. Introduction to Automation–

Industrial automation uses control systems to handle processes and machinery, replacing human labor for repetitive or hazardous tasks. The 'Automation Pyramid' defines the hierarchy from field devices to enterprise management. Automation can be fixed (hard-wired), programmable (PLC), or flexible (Robotics).

Study and application method

Sketch the 'Automation Pyramid' and label the levels; compare 'Fixed' and 'Programmable' automation conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'Automation Pyramid' and label the levels; compare 'Fixed' and 'Programmable' automation conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure തയ്യാറാക്കുക. ഏതു result ന്നുള്ള application തയ്യാറാക്കുക. Technical terms English-ൽ തയ്യാറാക്കുക.

Common mistake / precaution: Automation can lead to unexpected machine movements. Do not enter automated zones without authorised safety interlocks and adherence to lockout-tagout (LOTO) procedures.

2. Industrial Sensors and Actuators–

Sensors provide feedback from the process. Proximity sensors detect object presence without contact. LVDTs measure displacement. Actuators like pneumatic cylinders and hydraulic motors provide the mechanical force for movement, controlled by solenoid valves.

Study and application method

Describe the working of an 'Inductive Proximity Sensor'; explain the advantage of 'Pneumatic' actuators in hazardous environments.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of an 'Inductive Proximity Sensor'; explain the advantage of 'Pneumatic' actuators in hazardous environments.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result ന്നുള്ള application ന്നുള്ള Technical terms English-ൽ ന്നുള്ള.

Common mistake / precaution: Hydraulic systems operate under high pressure. Do not perform maintenance on pressurized lines without authorised pressure relief and safety shielding.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Programmable Logic Controllers (PLC)

PLC Architecture: CPU, Memory, I/O modules; Scan cycle and operating principle; Programming languages: Ladder logic, FBD; Logic gates, Timers, Counters in ladder logic.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

Identify core terms, explain the principle and complete the stated application or practical

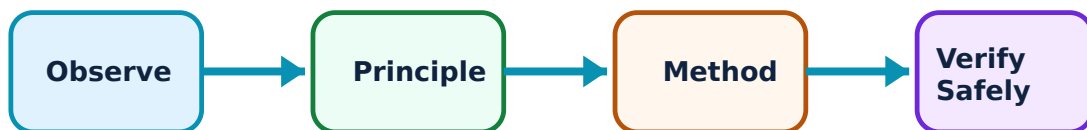
outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Programmable Logic Controllers (PLC): identify the system, state its principle, follow the correct method and verify the result safely.

1. PLC Architecture and Operation–

A PLC is a ruggedized computer for industrial control. It consists of a CPU, power supply, and I/O modules (Digital/Analog). The PLC operates in a 'Scan Cycle': Read Inputs -> Execute Program -> Update Outputs -> Housekeeping.

Study and application method

Sketch the 'Block Diagram' of a PLC; explain the 'Scan Cycle' and its impact on response time.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'Block Diagram' of a PLC; explain the 'Scan Cycle' and its impact on response time.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result ന്നുള്ള application ന്നുള്ള Technical terms English-ൽ എഴുതേണ്ടത്.

Common mistake / precaution: PLC programs can have logic errors. All code must be verified in a 'Simulation' or 'Test' mode before applying to actual hardware to prevent equipment damage.

2. Ladder Logic Programming–

Ladder logic is a graphical language resembling electrical relay circuits. It uses contacts (NO/NC) and coils. Timers (TON, TOF) and Counters (CTU, CTD) are used for time-based and event-based control. Comparison and math instructions allow for complex logic.

Study and application method

Write a ladder logic program for a 'Start-Stop' motor control with interlocking; explain the function of an 'On-delay Timer' (TON).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write a ladder logic program for a 'Start-Stop' motor control with interlocking; explain the function of an 'On-delay Timer' (TON).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നോക്കുക. ഏതു diagram / circuit / formula / procedure നോക്കുക. ഏതു result application നോക്കുക. Technical terms English-ൽ നോക്കുക.

Common mistake / precaution: Improper interlocking in ladder logic can cause short circuits or mechanical crashes. Always use authorised hardware interlocks in addition to software-based logic for safety-critical tasks.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

SCADA and Human Machine Interface (HMI)

SCADA architecture and features: Data acquisition, GUI, Alarms, Trends, Logging; HMI design and operation; Interfacing SCADA with PLC; Remote monitoring basics.

Approx. 16 hours

CO3

Understand · Apply

Learning goals

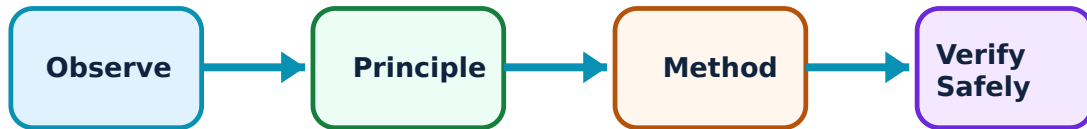
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for SCADA and Human Machine Interface (HMI): identify the system, state its principle, follow the correct method and verify the result safely.

1. SCADA Systems–

SCADA (Supervisory Control and Data Acquisition) provides high-level supervision of processes. It collects data from PLCs/RTUs and presents it to operators via a Graphical User Interface (GUI). Features include real-time alarms, historical trending, and data logging for compliance.

Study and application method

List four key features of a 'SCADA' system; explain the difference between 'Real-time' and 'Historical' trends.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List four key features of a 'SCADA' system; explain the difference between 'Real-time' and 'Historical' trends.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഡയഗ്രാം / സർക്യൂട്ട് / ഫോർമുല / പ്രോസീഡർ വ്യക്തമായി വിശദീകരിക്കുക. ഫോർമുലയിൽ റിസൾട്ട് വ്യക്തമായി വിശദീകരിക്കുക. Technical terms English-ൽ വ്യക്തമായി വിശദീകരിക്കുക.

Common mistake / precaution: SCADA systems are vulnerable to cyber-attacks. All industrial networks must follow authorised cybersecurity protocols, including firewalls and access control.

2. HMI and Operator Interaction–

An HMI (Human Machine Interface) is a local terminal (touchscreen or panel) that allows operators to interact with a specific machine or process. Good HMI design focuses on clarity, ease of use, and quick response to alarms.

Study and application method

Describe the role of an 'HMI' in a local automation cell; explain the importance of 'Alarm Management' in HMI design.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of an 'HMI' in a local automation cell; explain the importance of 'Alarm Management' in HMI design.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകുന്നു എന്ന് പറയുക. ഏതു diagram / circuit / formula / procedure നൽകുന്നു. ഏതു result നൽകുന്നു. application നൽകുന്നു. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Poor HMI design can lead to operator errors. All HMI layouts must follow authorised ergonomic and safety standards to ensure critical information is clearly visible.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

DCS and Communication Protocols

Distributed Control Systems (DCS) architecture; PLC vs DCS; Industrial communication protocols: Modbus, Profibus, CAN, Ethernet/IP; Industry 4.0 and IIoT overview.

Approx. 14 hours

CO4

Understand · Apply

Learning goals

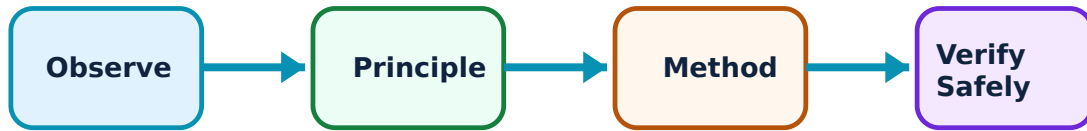
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for DCS and Communication Protocols: identify the system, state its principle, follow the correct method and verify the result safely.

1. Distributed Control Systems (DCS)–

DCS is used for large, complex, and continuous processes (e.g., oil refineries). Unlike a PLC-centric system, control is distributed among multiple controllers, improving reliability and scalability. DCS typically has integrated SCADA and advanced control functions.

Study and application method

Compare 'PLC' and 'DCS' in a conceptual table; describe the 'Redundancy' concept in DCS.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'PLC' and 'DCS' in a conceptual table; describe the 'Redundancy' concept in DCS.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure തയ്യാറാക്കുക. ഏതു result ന്നുള്ള application തയ്യാറാക്കുക. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: DCS failures can affect entire plants. All DCS installations must include authorised redundant power supplies and controllers for high availability.

2. Industrial Communication and Industry 4.0–

Industrial protocols allow devices to exchange data. Modbus is a simple serial protocol. Profibus and CAN are used for high-speed fieldbus communication. Industry 4.0 integrates automation with cloud computing, big data, and IIoT (Industrial Internet of Things) for smart manufacturing.

Study and application method

Explain the purpose of 'Modbus' protocol; list three components of 'Industry 4.0' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the purpose of 'Modbus' protocol; list three components of 'Industry 4.0' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result ന്നുള്ള application ന്നുള്ള Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Interoperability issues can arise between different protocols. All network integrations must follow authorised communication standards and gateway configurations.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Automation Fundamentals and Sensors

Use the concept flow to revise observation, principle, method and verification.

Module 2: Programmable Logic Controllers (PLC)

Use the concept flow to revise observation, principle, method and verification.

Module 3: SCADA and Human Machine Interface (HMI)

Use the concept flow to revise observation, principle, method and verification.

Module 4: DCS and Communication Protocols

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Write a ladder logic program to control a simple 'Traffic Light' sequence.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Develop a SCADA screen layout conceptually for a 'Water Tank' level control system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the block diagram of a DCS showing the operator station, engineering station, and field controllers.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the scan time conceptually for a PLC with a given number of instructions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify an automated system in a local factory or a video demonstration (e.g., bottling plant) and note its PLC type.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Automation Fundamentals and Sensors

Explain Introduction to Automation and state one correct method, application or precaution.—

Industrial automation uses control systems to handle processes and machinery, replacing human labor for repetitive or hazardous tasks. The 'Automation Pyramid' defines the hierarchy from field devices to enterprise management. Automation can be fixed (hard-wired), programmable (PLC), or flexible (Robotics).

Answer framework: Sketch the 'Automation Pyramid' and label the levels; compare 'Fixed' and 'Programmable' automation conceptually.

Important point: Automation can lead to unexpected machine movements. Do not enter automated zones without authorised safety interlocks and adherence to lockout-tagout (LOTO) procedures.

Explain Industrial Sensors and Actuators and state one correct method, application or precaution.—

Sensors provide feedback from the process. Proximity sensors detect object presence without contact. LVDTs measure displacement. Actuators like pneumatic cylinders and hydraulic motors provide the mechanical force for movement, controlled by solenoid valves.

Answer framework: Describe the working of an 'Inductive Proximity Sensor'; explain the advantage of 'Pneumatic' actuators in hazardous environments.

Important point: Hydraulic systems operate under high pressure. Do not perform maintenance on pressurized lines without authorised pressure relief and safety shielding.

Module 2 — Programmable Logic Controllers (PLC)

Explain PLC Architecture and Operation and state one correct method, application or precaution.—

A PLC is a ruggedized computer for industrial control. It consists of a CPU, power supply, and I/O modules (Digital/Analog). The PLC operates in a 'Scan Cycle': Read Inputs -> Execute Program -> Update Outputs -> Housekeeping.

Answer framework: Sketch the 'Block Diagram' of a PLC; explain the 'Scan Cycle' and its impact on response time.

Important point: PLC programs can have logic errors. All code must be verified in a 'Simulation' or 'Test' mode before applying to actual hardware to prevent equipment damage.

Explain Ladder Logic Programming and state one correct method, application or precaution.—

Ladder logic is a graphical language resembling electrical relay circuits. It uses contacts (NO/NC) and coils. Timers (TON, TOF) and Counters (CTU, CTD) are used for time-based and event-based control. Comparison and math instructions allow for complex logic.

Answer framework: Write a ladder logic program for a 'Start-Stop' motor control with interlocking; explain the function of an 'On-delay Timer' (TON).

Important point: Improper interlocking in ladder logic can cause short circuits or mechanical crashes. Always use authorised hardware interlocks in addition to software-based logic for safety-critical tasks.

Module 3 — SCADA and Human Machine Interface (HMI)

Explain SCADA Systems and state one correct method, application or precaution.—

SCADA (Supervisory Control and Data Acquisition) provides high-level supervision of processes. It collects data from PLCs/RTUs and presents it to operators via a Graphical User Interface (GUI). Features include real-time alarms, historical trending, and data logging for compliance.

Answer framework: List four key features of a 'SCADA' system; explain the difference between 'Real-time' and 'Historical' trends.

Important point: SCADA systems are vulnerable to cyber-attacks. All industrial networks must follow authorised cybersecurity protocols, including firewalls and access control.

Explain HMI and Operator Interaction and state one correct method, application or precaution.—

An HMI (Human Machine Interface) is a local terminal (touchscreen or panel) that allows operators to interact with a specific machine or process. Good HMI design focuses on clarity, ease of use, and quick response to alarms.

Answer framework: Describe the role of an 'HMI' in a local automation cell; explain the importance of 'Alarm Management' in HMI design.

Important point: Poor HMI design can lead to operator errors. All HMI layouts must follow authorised ergonomic and safety standards to ensure critical information is clearly visible.

Module 4 — DCS and Communication Protocols

Explain Distributed Control Systems (DCS) and state one correct method, application or precaution. –

DCS is used for large, complex, and continuous processes (e.g., oil refineries). Unlike a PLC-centric system, control is distributed among multiple controllers, improving reliability and scalability. DCS typically has integrated SCADA and advanced control functions.

Answer framework: Compare 'PLC' and 'DCS' in a conceptual table; describe the 'Redundancy' concept in DCS.

Important point: DCS failures can affect entire plants. All DCS installations must include authorised redundant power supplies and controllers for high availability.

Explain Industrial Communication and Industry 4.0 and state one correct method, application or precaution. –

Industrial protocols allow devices to exchange data. Modbus is a simple serial protocol. Profibus and CAN are used for high-speed fieldbus communication. Industry 4.0 integrates automation with cloud computing, big data, and IIoT (Industrial Internet of Things) for smart manufacturing.

Answer framework: Explain the purpose of 'Modbus' protocol; list three components of 'Industry 4.0' conceptually.

Important point: Interoperability issues can arise between different protocols. All network integrations must follow authorised communication standards and gateway configurations.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Automation Fundamentals and Sensors.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Programmable Logic Controllers (PLC).
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: SCADA and Human Machine Interface (HMI).
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: DCS and Communication Protocols.

8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition and hierarchy of industrial automation (Automation Pyramid); Fixed, Programmable, and Flexible automation; Industrial sensors: Proximity, LVDT, Pressure, Flow; Actuators: Pneumatic and Hydraulic..
2. Develop a complete answer or practical plan for: PLC Architecture: CPU, Memory, I/O modules; Scan cycle and operating principle; Programming languages: Ladder logic, FBD; Logic gates, Timers, Counters in ladder logic..
3. Develop a complete answer or practical plan for: SCADA architecture and features: Data acquisition, GUI, Alarms, Trends, Logging; HMI design and operation; Interfacing SCADA with PLC; Remote monitoring basics..
4. Develop a complete answer or practical plan for: Distributed Control Systems (DCS) architecture; PLC vs DCS; Industrial communication protocols: Modbus, Profibus, CAN, Ethernet/IP; Industry 4.0 and IIoT overview..

LAST-MILE REVISION

Quick Revision

Module 1: Automation Fundamentals and Sensors

Definition and hierarchy of industrial automation (Automation Pyramid); Fixed, Programmable, and Flexible automation; Industrial sensors: Proximity, LVDT, Pressure, Flow; Actuators: Pneumatic and Hydraulic.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Programmable Logic Controllers (PLC)

PLC Architecture: CPU, Memory, I/O modules; Scan cycle and operating principle; Programming languages: Ladder logic, FBD; Logic gates, Timers, Counters in ladder logic.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: SCADA and Human Machine Interface (HMI)

SCADA architecture and features: Data acquisition, GUI, Alarms, Trends, Logging; HMI design and operation; Interfacing SCADA with PLC; Remote monitoring basics.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: DCS and Communication Protocols

Distributed Control Systems (DCS) architecture; PLC vs DCS; Industrial communication protocols: Modbus, Profibus, CAN, Ethernet/IP; Industry 4.0 and IIoT overview.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Introduction to Automation	Industrial automation uses control systems to handle processes and machinery, replacing human labor for repetitive or hazardous tasks.
Industrial Sensors and Actuators	Sensors provide feedback from the process.
PLC Architecture and Operation	A PLC is a ruggedized computer for industrial control.
Ladder Logic Programming	Ladder logic is a graphical language resembling electrical relay circuits.
SCADA Systems	SCADA (Supervisory Control and Data Acquisition) provides high-level supervision of processes.
HMI and Operator Interaction	An HMI (Human Machine Interface) is a local terminal (touchscreen or panel) that allows operators to interact with a specific machine or process.
Distributed Control Systems (DCS)	DCS is used for large, complex, and continuous processes (e.
Industrial Communication and Industry 4.0	Industrial protocols allow devices to exchange data.

LEARNING RESOURCES

References

Recommended industrial automation references

1. Frank D. Petruzella, Programmable Logic Controllers.
2. Gary Dunning, Introduction to Programmable Logic Controllers.
3. Stuart A. Boyer, SCADA: Supervisory Control and Data Acquisition.
4. Krishna Kant, Computer Based Industrial Control.

Approved public industrial automation sources

- <https://nptel.ac.in/courses/108105062>
- <https://nptel.ac.in/courses/108105088>

These sources provide the verified study basis while official Course 6044C detail is unavailable; they do not replace official plant designs, authorised safety instrumented systems, certified equipment specifications or authorised production plans.